Kinematic Modelling and MATLAB-based Analysis of the Dobot Magician Lite Robot Arm

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Abstract

This project develops and validates a MATLAB implementation of forward kinematics, inverse kinematics, and Jacobian analysis for the Dobot Magician Lite robot arm. Using a 4-degree-of-freedom model with revolute joints and a simple parallelogram mechanism, the study derives the forward and inverse kinematics, computes the Jacobian, and evaluates singularities and joint limits. The Dobot Magician Lite is a desktop serial manipulator consisting of four revolute joints. The forward kinematics are derived using standard DH parameters, while the inverse kinematics are solved using a geometric decomposition into a planar two-link problem for both joint and Cartesian coordinate systems, enabling flexible motion control and workspace analysis. The Jacobian is used to study singularities and joint limits, which are important for safe operation and for understanding how small joint errors propagate, making it suitable for both academic and practical applications [8].

As a classic robotics perspective, the Dobot Magician Lite can be compared to a serial articulated robot arm. Compared to parallel manipulators, serial manipulators offer a larger workspace and greater flexibility, but may suffer from lower stiffness and accumulated positioning errors. Compared to SCARA robots, the Dobot Magician Lite has more vertical flexibility but may have reduced speed and precision in planar tasks.

The Dobot Magician Lite is particularly suitable for educational use because it is low-cost, compact, and supported by computer-friendly tools [9]. However, limitations include restricted payload capacity (250 g), limited reach (340 mm), and lower repeatability (±0.2 mm) compared to high-end robotic systems [1]. The aim of this project was to develop an analytical and MATLAB-based model of the Dobot Magician Lite, to derive its forward and inverse kinematics, inverse velocity kinematics, Jacobian, singularities and joint limits, and to validate the behaviour.

List of Symbols

θ_1	base rotation angle
θ_2	shoulder joint angle
θ_3	elbow joint angle
θ_4	end-effector rotation angle
L_1, L_2, L_3	main arm link lengths

1 Kinematic Modelling

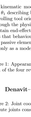
1.1 Robot configuration and assumptions

The kinematic model therefore treats the robot as a 4R serial manipulator, with the fourth joint used to orient the end-effector independently of the three-link planar chain.

Although the physical robot includes a parallelogram mechanism [10] that keeps the end-effector orientation, the kinematic model does not need to represent this mechanism because it maintains an equivalent passive transformation ($-\theta_2 - \theta_3$). The passive joints are treated as a mechanical feature of the robot, not as additional degrees of freedom, but only as a modelling device used to preserve the geometric constraint.

Figure 1: Appearance of the Dobot Magician Lite showing the physical arrangement of the four revolute joints used in the kinematic model. Adapted from [8]

Figure 2: Diagram of the Dobot Magician Lite showing the physical arrangement of the four revolute joints used in the kinematic model. Adapted from [8]



The Denavit-Hartenberg parameters derived from the frame assignment in section 1.2 are summarised in Table 1.

Table 1: Denavit-Hartenberg parameter table derived from the frame assignment shown in Figure 3.

Link	a_i (mm)	α_i	d_i (mm)	θ_i
1 - Base	0	$90°$	0	θ_1
2 - Rear Arm	150	0	0	θ_2
3 - Forearm	160	0	0	θ_3
4 - parallelogram link	0	0	0	$-\theta_2 - \theta_3$
5 - End-effector	0	0	0	θ_4

The coordinate frames were assigned according to the standard Denavit-Hartenberg convention [11]. Each axis was aligned with the joint axis of rotation of the corresponding joint, while each x-axis was defined along the common normal between consecutive joint axes. The parallelogram mechanism was treated as a passive constraint rather than as additional degrees of freedom. Its effect on the end-effector orientation approximately cancels the shoulder and elbow rotations, so the end-effector orientation approximately depends only on the base rotation θ_1 and the end-effector rotation θ_4.

Figure 3: Denavit-Hartenberg frame assignment used for the kinematic modelling of the Dobot Magician Lite manipulator.

The Denavit-Hartenberg parameter table derived from the frame assignment is shown in Table 1.

1.2 Forward kinematics

Forward kinematics for the Dobot Magician Lite were derived using the Denavit-Hartenberg convention. A homogeneous transformation matrix was assigned to each link, in order to obtain the position and orientation of each link frame relative to the preceding frame. The overall transformation from the base to the end-effector was then obtained by multiplying these transformation matrices sequentially [2, 3].

The forward kinematics model used in this project, the total transformation is written as

$$T_{04} = A_1 A_2 A_3 A_4 A_5$$

where A_i denotes the homogeneous transformation matrix associated with the i-th link, and T_{04} is the end-effector pose. The passive joints is included only to preserve the geometric behaviour of the mechanism and do not represent an additional actuated degree of freedom.

The forward kinematics transformation matrix provides both the orientation of the end-effector and its Cartesian position relative to the base frame. The position components are extracted from the last column of the overall transformation equation and used to construct the Jacobian and to evaluate the reachable workspace. The orientation component is used to check joint limits and singularities.

1.3 Jacobian and singularity analysis

The Jacobian matrix relates joint rates to end-effector motion. In this work, the geometric Jacobian was used to derive the relationship between joint velocities and the linear and angular velocity of the end-effector, while an analytical form was considered for simplicity and speed. The full derivation of the analytical expressions and to construct the Jacobian is particularly useful for identifying singularities which the singularities lower mobility or becomes difficult to control.

Geometric Jacobian

The geometric Jacobian relates the joint velocities to the linear and angular velocity of the end-effector:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = J_g(q)\dot{q}$$

The geometric Jacobian is written as $\dot{\mathcal{L}} = J_g(q)\dot{q}$ and is given by

$$J_g(q) = \begin{bmatrix} -r\sin(\theta_1) & -r\cos(\theta_1)[-L_2\sin(\theta_2) - L_3\sin(\theta_2 + \theta_3)] & -L_3\sin(\theta_2+\theta_3)\cos(\theta_1) & 0 \\ r\cos(\theta_1) & -r\sin(\theta_1)[-L_2\sin(\theta_2) - L_3\sin(\theta_2 + \theta_3)] & -L_3\sin(\theta_2+\theta_3)\sin(\theta_1) & 0 \\ 0 & L_2\cos(\theta_2) + L_3\cos(\theta_2+\theta_3) & L_3\cos(\theta_2+\theta_3) & 0 \\ 0 & -\sin(\theta_1) & -\sin(\theta_1) & 0 \\ 0 & \cos(\theta_1) & \cos(\theta_1) & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Where: θ_1 is the base rotation, θ_2 is the shoulder joint angle, θ_3 is the elbow joint angle, θ_4 is the tool rotation, L_2 is the rear arm length, L_3 is the forearm length, $r = L_2\cos(\theta_2) + L_3\cos(\theta_2 + \theta_3)$ is the radial reach in the Jacobian because it is a constant term that does not affect the velocity mapping.

Analytical Jacobian

The analytical Jacobian satisfies $\dot{x} = J_a(q)\dot{q}$ and is given by

$$J_a(q) = \begin{bmatrix} -r\sin(\theta_1) & -r\cos(\theta_1)[-L_2\sin(\theta_2) - L_3\sin(\theta_2 + \theta_3)] & -L_3\sin(\theta_2+\theta_3)\cos(\theta_1) & 0 \\ r\cos(\theta_1) & -r\sin(\theta_1)[-L_2\sin(\theta_2) - L_3\sin(\theta_2 + \theta_3)] & -L_3\sin(\theta_2+\theta_3)\sin(\theta_1) & 0 \\ 0 & L_2\cos(\theta_2) + L_3\cos(\theta_2+\theta_3) & L_3\cos(\theta_2+\theta_3) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The base offset b appears in the forward kinematics of x, but it does not appear in the Jacobian because it is a constant term. Its derivative with respect to all joint variables is zero.

Singularities

A singularity occurs when the determinant of the analytical Jacobian becomes zero:

$$\det(J_a) = L_3 L_2 r \sin(\theta_3)$$

This expression shows that singularities occur when either $\sin(\theta_3) = 0$ or $r = 0$. The condition $\sin(\theta_3) = 0$ corresponds to the arm being fully extended or fully folded, so that the links become collinear. The condition $r = 0$ corresponds to the wrist lying on the base axis, where the manipulator loses motion authority in certain horizontal directions. Near configurations, the velocity mapping becomes ill-conditioned and large joint velocities may be required to generate small end-effector motion [12].

1.5 Inverse kinematics

The inverse kinematics problem was solved using a geometric approach, because it provides closed-form solutions and is well suited to two-link planar arm structures [2, 5].

Figure 4: Top-view geometry used to compute the base joint angle θ_1.

The base joint angle was obtained from the projection of the target position:

$$\theta_1 = \operatorname{atan2}(y, x)$$

The radial distance in the horizontal plane is $r = \sqrt{x^2 + y^2}$. The remaining problem is reduced to a planar two-link geometry in the r - z plane [2]. A geometric inverse kinematics model was selected because it provides closed-form solutions and is well suited to two-link planar arm structures [2, 5].

Figure 5: Planar geometric construction used for the inverse kinematics derivation.

Figure 5 shows the planar geometry used to derive the shoulder and elbow solutions. Two solutions branches exist for reachable targets: an elbow-up configuration and an elbow-down configuration. Accordingly, the shoulder angle can be written as

$$\theta_2 = \alpha \pm \beta$$

for one branch and

$$\theta_2 = \alpha - \beta$$

for the other. The elbow angle θ_3 is then obtained from the corresponding trigonometric relations, giving two valid inverse-kinematics solutions for the same target position.

3 Materials and Methods

3.1 Kinematic Modelling Approach

The kinematic model of the Dobot Magician Lite was developed using the Denavit-Hartenberg (DH) convention, which provides a standard representation for describing the spatial relationship between adjacent links. Each joint was modelled as a revolute joint with one rotational degree of freedom. For each kinematic chain, despite the physical presence of a four-bar linkage mechanism, a passive link was included in the model to maintain the constraint that the end-effector remains parallel to the ground. This simplification allows the kinematic model to remain tractable while still representing the mechanism's relevant kinematic behaviour.

3.2 Inverse Kinematics Strategy

A geometric approach was selected for solving the inverse kinematics problem. This method was chosen due to the relatively simple structure of the robot and the availability of clear geometric relationships between joint angles and end-effector position.

The inverse kinematics solution was derived by:

- First solving for the base rotation $\theta_1 = \operatorname{atan2}(y, x)$
- Reducing the 3D problem into a 2D planar problem in the (r, z) plane
- Applying trigonometric relationships and cosine law to determine θ_2 and θ_3
- Considering both elbow-up and elbow-down configurations

This approach was preferred over numerical methods due to its computational efficiency, analytical clarity, and reliability for real-time applications.

3.3 Dynamic modelling approach

The dynamic behaviour of the manipulator was considered using the standard rigid-body robot dynamics formulation [3]

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q)$$

where τ is the vector of joint torques, $M(q)$ is the inertia matrix, $C(q, \dot{q})\dot{q}$ represents Coriolis and centrifugal effects, and $G(q)$ is the gravity vector.

In the present project, the dynamic equations were incorporated at a conceptual level to support the overall modelling framework and to identify τ in the future. However, a complete Newton-Euler implementation was limited by the lack of detailed physical parameters such as link masses, mass centres, and inertia tensors from the manufacturer documentation [4]. Therefore, the work focused on establishing the dynamic form of the equations and explaining how such a term would be integrated into the MATLAB model for future extension.

3.4 MATLAB Implementation

The robot model was implemented in MATLAB using a modular approach. Separate functions were created for Denavit-Hartenberg transformations, forward kinematics evaluation, inverse kinematics solution, Jacobian computation, and singularity analysis. This structure allows individual components to be tested and improved without affecting the overall system. Intermediate results which were then evaluated numerically during testing [5].

The implementation was designed so that a complete kinematic model could be provided as input, the corresponding joint angles could be computed through inverse kinematics, and the resulting end-effector pose could then be verified through forward kinematics reconstruction. Additional checks were included to detect unreachable targets and singular configurations.

3.5 Testing Framework

A structured testing framework was used to validate the correctness of the implemented kinematics. Forward kinematics results were compared against known reference poses, while inverse joint configurations were tested by computing end-effector positions with the expected geometric outcome. Inverse kinematics solutions were then fed back into forward kinematics to reconstruct the end-effector position with the expected geometric outcome. Inverse kinematics was considered valid when the reconstructed position matched the original target and substituting the solution back into the forward kinematics equations.

Additional tests were used to verify singularity detection and workspace rejection for unreachable targets. Together, these checks ensured internal consistency between the analytical formulation and the computational implementation.

4 Results and Validation

The MATLAB model was validated using forward-kinematics, inverse-kinematics, and singularity test cases. These results were used to verify the implemented equations and identify discrepancies with expected geometric outcomes, while inverse kinematics was verified by reconstructing the end-effector position in Cartesian positions. Additional tests were used to confirm singularity detection and rejection of unreachable targets.

Table 2: Validation of forward and inverse kinematics for representative test cases.

Test case	Input	Output
Forward kinematics	$q = (0.5236, 0.7854, 0.5)$ rad	$(x, y, z) = (0.1837, 0.106, 0.296)$
Inverse kinematics	Target $(0.2000, 0.0000, 0.2000)$	$\theta_1 = 0.0000$ rad, $\theta_2 = 0.7854$ rad, $\theta_3 = 0$
Singularity test	$\theta_3 = 0$	$\det(J) = 0$
Workspace test	Unreachable valid; invalid cases	Correctly rejected

The validation results show good agreement between the analytical model and MATLAB implementation. In particular, the forward kinematics matched the hand kinematics output and the inverse kinematics reconstruction confirmed that the position targets. In addition, the singularity and workspace checks behaved as intended, with the implementation correctly rejecting unreachable targets within the robot workspace.

Figure 6: Successful inverse kinematics solution using the MATLAB-based UI.

Figure 7: Detection of unreachable target in inverse kinematics.

The system correctly identifies when a target position lies outside the robot's workspace. In this example, the requested distance exceeds the maximum reach of the robot ($L_2 + L_3 = 0.30$ m), resulting in an error message. This demonstrates the robustness of the inverse kinematics implementation: proper workspace validation and prevents physically invalid solutions.

5 Discussion and Conclusion

This project developed and validated a kinematic model of the Dobot Magician Lite using the Denavit-Hartenberg formulation and implemented the main equations in MATLAB. Forward and inverse kinematics were derived analytically, the Jacobian was obtained for differential motion analysis, and singular configurations were identified from the analytical form of the Jacobian. The implementation was able to reproduce valid end-effector positions and reject unreachable targets within the robot workspace.

One of the main difficulties was correctly representing the parallelogram mechanism of the robot within a tractable analytical framework. This was addressed by treating the passive joints as constraints rather than additional degrees of freedom, since accurate link masses, centres of mass and inertia tensors were not available to provide a sufficient detail from the manufacturer documentation.

Despite these limitations, the final model provides a robust analytical and computational framework for studying the motion of the Dobot Magician Lite. The results demonstrate that the model is suitable for forward kinematics, inverse kinematics, Jacobian analysis, and singularity detection. The work also shows that a simplified model can be used for incorporating more detailed dynamic parameters, actuator constraints, joint limits and closed-loop control in order to improve realism and practical applicability.

Figure 8: Principal dimensions of the Dobot Magician Lite used to support the geometric assumptions of the model. Adapted from [10]

6 References

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